Step 4c: Structural Response Calculation and Evaluation under Prescribed Transient Loads

Detailed Hierarchical Methods Comparison Review Based on the Step4c Exact146 Corpus

PETW Manuscript

Expanded English literature review

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Extended review based on the Step 4c exact146 literature library: response computation, metric systems, and uncertainty boundaries following the prescription of downburst or transient wind loads

Extended English Literature Review

Evidence and use boundary

This review draws on 146 controlled accepted sources as its evidence universe and is used to organize research methods, metric definitions, and uncertainty boundaries for structural response computation and evaluation under prescribed transient loads. Statements herein do not constitute a PETW structural-response computation result, do not form design coefficients, code-adoption conclusions, or engineering adoption decisions. Scientific status is `not_approved`; formal manuscript status is `not_applied`. No structural-response computation, Zotero write, or formal main-manuscript application has been executed. All numerical values remain local to the source structure, loading configuration, solution method, and validation scope; cross-source quantitative derivation is not performed until comparability conditions have been verified.

Abstract

This review uses 146 accepted sources as a controlled evidence universe and systematically organizes research on structural response computation and evaluation following the prescription of downburst or other transient wind loads. Starting from the problem boundary of the transient loading chain, the review first analyzes how thunderstorm wind-field elements enter the interface stages of structural response computation, then provides a dedicated comparison of the wind-speed-to-pressure-or-force relationship. Information retention, validation basis, cost, and uncertainty serve as comparison axes for the wind-tunnel, CFD, and theoretical-analytical route families. From the pressure-to-structural-degrees-of-freedom interface layer, the review discusses linear dynamic response methods (including time-domain, frequency-domain or evolutionary-spectral, time-frequency, and response-spectrum routes) and nonlinear, aerodynamic-feedback, and failure-mechanism-oriented responses. Response evidence across structural archetypes is treated in three groups: buildings and cladding, tower-conductor systems, and shells, long-span structures, and slender structures, with each group retaining the source-specific inputs, methods, and response metrics. The discussion of the response-metric system focuses on the definitional ambiguity and window-length dependence of peak values, root-mean-square values, dynamic amplification factors, peak factors, and equivalent static demands under transient conditions. Uncertainty propagation is tracked stage by stage through the variability sources in inflow characterization, pressure normalization, load mapping, damping and frequency, nonlinear paths, and response extraction. The review terminates at the organized output of response evidence, metric definitions, sensitivity information, and use limitations, without entering design-value derivation or code-adoption conclusions.

Keywords: transient wind loads; structural response; nonstationary dynamics; downburst; response metrics; uncertainty propagation

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1 Problem Boundary and Position in the Loading Chain

1.1 Division of labor among wind field, pressure/force, structural response, and design load

The chain of transient wind action on structures can be divided into four stages: wind-field characterization, surface pressure or force generation, structural dynamic response, and response metrics with use boundaries. Physical simulation techniques for nonstationary strong wind events such as downbursts have evolved from gravity currents to impinging jets to multi-fan actively controlled wind tunnels, with each generation exhibiting significant differences in nozzle diameter, scale ratio, and controllable degrees of freedom [3]. Stand-alone simulators are limited by fewer controllable parameters, whereas combined simulators achieve higher control over nonstationary features by embedding active devices within boundary-layer wind tunnels. When the wind field or equivalent load has been delivered to the structural analyst, the problem is not thereby well-posed: numerical simulation of wind effects itself involves multiple uncertainty sources from a probabilistic perspective, from inflow turbulence properties to the choice of pressure-normalization reference, all of which influence the final response estimate [65], and the limited number of observed events further constrains the reliability of the probabilistic description.

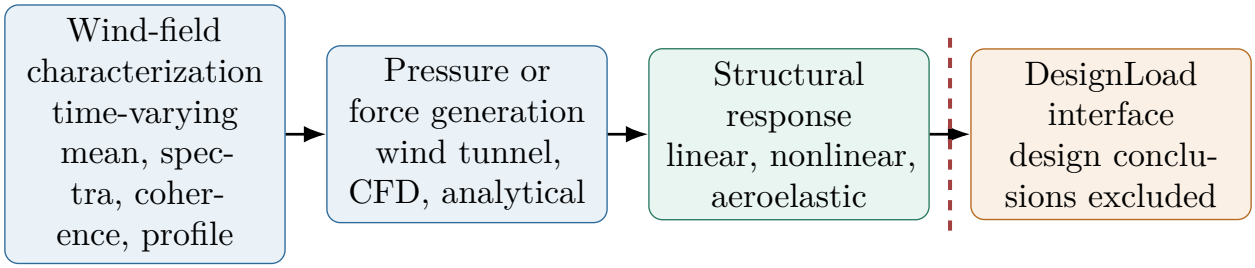


Figure 1 Schematic of the transient wind-action loading chain and division boundaries. Original synthesis based on exact146 accepted-only evidence; does not redraw source numerical curves and does not represent a PETW computation result. From left to right: wind-field characterization (WindField interface scope), pressure or force generation (WindTunnel/CFD and analytical routes), structural dynamic response (Structural Response, the scope of this review), and design load or code interface (DesignLoad boundary, outside the scope of this review). Arrows indicate the direction of information flow; the vertical boundary line marks the boundary of design-value derivation, where this review terminates.

This review focuses on the third stage of the chain: solving the structural dynamic response after the transient load has been prescribed, constructing metrics, and delineating evidence boundaries. The starting point is the completed delivery of the pressure field or equivalent forces; the endpoint is the organized output of response quantities, metric definitions, and use limitations.

1.2 Hierarchy of input objects: wind speed, pressure, base force, nodal force, generalized force, equivalent static force

The representation form of the prescribed load itself constitutes the first level of ambiguity. Loads can be delivered as pressure fields, aerodynamic force coefficients, base forces, or generalized forces, and different forms retain different amounts of spatial information, thereby imposing different constraints on subsequent response computation. For transmission-line systems, the evaluation of downburst loading involves adaptation issues across stages from the wind-field model to the tower-conductor system [16], and review literature has organized the hierarchical relationships among wind-field generation, load application, and structural response [56]. The ambiguity in load form can continue to propagate downstream even after the input representation has been determined: a pressure field contains full spatial-distribution information, whereas a base force is merely the spatial-integral result, and the two support fundamentally different response-analysis paths.

1.3 Evidence scope and non-approval boundary

The delineation of the structural scope determines the organization of response observables. A general loading-chain framework for nonstationary wind effects links time-frequency modeling to structural response analysis [138], but different structural archetypes exhibit markedly different response-observable systems: the global displacement and acceleration of buildings, the support reactions and conductor tensions of transmission towers, and the multi-modal participation components of long-span structures differ in physical standing and sensitive parameters. The complexity of three-dimensional nonstationary non-Gaussian wind-field characterization further amplifies the difficulty of cross-archetype comparison [109]; accordingly, this review treats three archetype groups in separate sections: buildings and cladding, tower-conductor systems, and shells, long-span structures, and slender structures.

1.3.1 Review endpoint and DesignLoad boundary

Information transfer between stages of the loading chain involves not only numerical magnitudes but also the transfer of comparability conditions. When a wind-tunnel test uses a specific reference dynamic pressure and scale ratio to obtain pressure coefficients, the physical meaning of those coefficients is confined to the apparatus conditions and the geometric features of the building model. When such pressure coefficients are used for structural response computation, supplementary information including scale ratio, reference-velocity definition, measurement precision, and frequency resolution must be transferred simultaneously. If such conditions are lost during transfer, the physical meaning of the subsequent response computation becomes uncertain.

The endpoint boundary of this review is set at response evidence, metric definitions, sensitivity rankings, and use limitations, without entering design-value derivation. Research on thunderstorm-wind design parameters indicates that an independent methodological chain exists between response evidence and design parameters [96], and performance-based design methods relying on historical records and Monte Carlo simulation likewise emphasize the logical separation between design-parameter generation and response analysis [105]. This review states, for each cited source, only the response evidence reported under its given configuration; all numerical values are confined to the source's structural object, loading conditions, and solution method, without deriving design-use coefficients or code-adoption conclusions from local evidence.

2 How Thunderstorm Wind-Field Elements Enter Structural Response Computation

2.1 Time-varying mean, moving path, and event duration

The first requirement for nonstationary wind-effect analysis is to separate each layer of the loading chain explicitly. The characterization of the time-varying mean flow, the modeling of the fluctuating component, the conversion to pressure or force, and the structural-dynamics solution must maintain independent definitions within the analytical framework; any implicit merging of layers invalidates subsequent comparability judgments. Review literature on downburst wind-field characteristics and structural wind effects organized the application of time-domain computation paths across different structural archetypes [28]; that review characterized the time-varying mean as a function of radial-velocity peak height varying with time and radial position, and identified the nose-shaped vertical profile as the key feature distinguishing downbursts from boundary-layer winds. The path of a moving downburst determines the effective load time history experienced by the structure: as the downburst center traverses the structural location at a given translation velocity, the structure sequentially experiences the ascending segment of the vortex front, the radial-velocity peak, and the tail decay segment, and the ascent rate and duration together determine the frequency content of the load time history.

A proper orthogonal decomposition study of two full-scale nonstationary downburst wind-speed vertical profiles [21] decomposed the time-domain signal into a time-varying mean, a time-varying variance, and normalized fluctuations, finding that at least four modes are needed to capture more than 95 percent of the energy of the normalized fluctuations, which indicates that simple time-varying-mean extraction cannot fully strip the nonstationary features. Research on practical methods for digital simulation of non-synoptic wind speeds and their implications for single-pole tower response [60] further demonstrated the influence of randomizing nose-shaped-profile parameters on the probabilistic description of structural response.

2.2 Variance, turbulence intensity, non-Gaussianity, and nonstationarity assumptions

Another key parameter of the moving path is the translation velocity of the downburst. The translation velocity alters the frequency content of the load perceived by the structure through a Doppler effect; it changes the effective load-action time and the ascent-segment morphology, and its influence on resonance accumulation and local overshoot depends on the ratio of event time scale to structural period and cannot be ranked independently of the source configuration. Analytical prediction of quasi-static response of transmission lines subjected to moving downbursts [27] demonstrated in a six-span case that the mean response accounts for more than 75 percent of the total response, indicating the dominance of the quasi-static component under low translation-velocity conditions. Research on the significance of convective gusts [48] noted from a meteorological perspective that gust durations in convective events are typically on the order of tens of seconds to several minutes, far shorter than the typical observation windows for boundary-layer winds; this time-scale difference constitutes a fundamental challenge for structural dynamic response analysis.

The duration of the transient event determines whether the structure can develop to its stationary resonance level. The ARMA-GARCH model for nonstationary wind-speed characterization demonstrated, from a time-series modeling perspective, the path for extracting time-varying variance; that model verified the self-similar properties of the normalized power spectral density on a measured thunderstorm record at 497 m height on the Shanghai World Financial Center, with its validity depending on the establishment conditions of the uniform modulation assumption [4]. When the rate of change of the time-varying variance is of the same order as the structural fundamental period, the build-up and decay of resonant components themselves become time-varying, and the classical stationary dynamic amplification factor is no longer applicable.

Research on the complexity of physical simulation for transient downburst-type events discussed the minimum requirements for simulation apparatus to reproduce time-varying variance characteristics, noting that not all physical simulations need to reproduce the complete transient process, but the minimum requirements should include controllability of the time-varying mean and spectral compatibility of the fluctuating component [45]. The distribution of turbulence intensity with height under nose-shaped-profile conditions differs systematically from the logarithmic-law distribution in boundary layers; this difference directly affects the relative magnitudes of the fluctuating loads received at each floor of the structure.

2.3 Spectra, evolutionary spectra, and time-frequency representations

Non-Gaussianity and nonstationarity often co-occur in downburst wind-speed records. Wind-tunnel measurements of a 3:2 rectangular cross-section recorded the co-existence of non-Gaussian and nonstationary features in pressure statistics [69], with skewness and kurtosis undergoing significant changes during the rapid wind-speed ramp-up phase. Research on three-dimensional nonstationary non-Gaussian wind fields and their implications for wind-structure interaction [109] emphasized the scarcity of measured records simultaneously satisfying all three characteristics of three-dimensionality, nonstationarity, and non-Gaussianity; this constitutes a fundamental data constraint for probabilistic analysis of structural response.

Evolutionary power spectral density provides a systematic description of how the frequency content of a nonstationary process varies with time. An evolutionary PSD model for thunderstorm outflows [74] was constructed under conditions consistent with full-scale records; that model separately fitted the temporal modulation function and the frequency-dependent function, with its parameters confined to specific measurement stations and event samples. Research on time-varying coherence functions based on evolutionary spectra [2] proposed two time-varying coherence models and fitted them to two downburst events recorded at the Texas Tech University tower; results showed that the spatial correlation structure of nonstationary downburst wind fields varies with time: when a single-modulation oscillatory-process assumption is used, the coherence function degenerates to a time-invariant form, whereas a more general sigma-oscillatory-process assumption preserves the time-varying character of the coherence function.

2.4 Coherence functions, spatial correlation, and generalized-force phase combination

Spatial correlation exhibits time-varying characteristics under nonstationary conditions, and its reduction effect on resultant forces cannot simply follow the coherence-decay models from stationary conditions. An experimental study on the aerodynamic coefficients and pressure distributions of two finite-height circular cylinders in downburst-type outflows [83] showed that, as radial position changes, the flow structure around the same cylinder transitions from attached flow to separated flow, and pressure correlation undergoes a transition from high to low correlation across different development stages of the outflow. A fast simulation method for nonstationary wind-speed fields based on proper-orthogonal-decomposition interpolation [97] improved computational efficiency by one order of magnitude relative to the direct spectral representation method while preserving the time-varying correlation structure at each spatial point.

Physical and numerical modeling of downburst gust fronts used two classes of schemes, density-difference-driven gravity currents and momentum-driven impinging jets [102]; the two reproduce systematically different leading-edge vortex structures and near-surface coherence characteristics: the gravity-current scheme more closely approximates atmospheric observations in head-vortex height and propagation speed, while the impinging-jet scheme offers greater controllability in near-surface maximum-velocity height. Experiments with a multi-blade flow-control device in a boundary-layer wind tunnel producing nonstationary wind-load effects [108] recorded an asymmetric pattern in the inter-sensor correlation coefficient between the acceleration and deceleration phases. These results imply that the direct use of spatial-correlation empirical formulas obtained from stationary winds in transient winds may introduce systematic bias.

2.5 Profile, reference height, and wind-direction variation

The shape of the vertical profile determines the relative weighting of loads at each height. The nose-shaped vertical profile of downbursts and the monotonically increasing profile of boundary-layer winds produce fundamental differences in moment-arm distributions and modal-load allocation. Boundary-layer wind-tunnel experimental simulation of thunderstorm profiles used passive grids to generate quasi-static nose-shaped wind-speed profiles [6]; such stationary devices produce load time histories that can serve as input baselines for structural analysis but lack the nonstationary temporal-modulation features. The choice of reference height affects the meaning of pressure coefficients and force coefficients: when the reference velocity changes from the peak height to the 10 m standard height or the rooftop height, the dimensionless coefficients corresponding to the same surface-pressure record change numerically.

Radial position and translation velocity further modulate the temporal features of the loads experienced by the structure. Numerical pressure analysis of a rectangular cross-section in a moving thunderstorm downburst wind field [39] showed that translation velocity alters the frequency of interaction between vortex structures and the cross-section. Wind-direction variation in downburst radial

outflow exhibits rotation or deflection characteristics, differing from the gradual height-dependent change in boundary-layer wind direction. Directional decomposition and characterization of thunderstorm outflows [118] extracted spectral parameters by direction from measured data; this decomposition allows the evolutionary spectra of the along-wind and cross-wind components to each maintain physical consistency.

2.6 Comparison of response influence: time-domain path versus time-frequency/frequency-domain path

From the perspective of structural response computation, the above wind-field elements enter response analysis through three paths. The time-domain path directly feeds the step-by-step wind-speed or load time history into the equation-of-motion solver, preserving the phase relationship and transient peaks between load and response. The frequency-domain or evolutionary-spectral path describes the nonstationary input using evolutionary power spectral density and obtains the second-order moment statistics of the response through a time-varying frequency-response function, but requires that the input process be expressible in a uniform-modulation or slowly-varying-modulation form. The time-frequency path simultaneously preserves time and frequency information in the wavelet-transform or S-transform domain, allowing tracking of instantaneous-frequency changes, but time-frequency resolution is constrained by the uncertainty principle.

The three paths entail different trade-offs in information retention, computational cost, and assumption stringency. FFT simulation of multivariate nonstationary random processes [70] addressed the generation of multi-point nonstationary signals in the frequency domain; its efficiency gives the frequency-domain path an advantage at the sample-generation stage. Analysis, modeling, and simulation of thunderstorm winds in two orthogonal horizontal directions [82] demonstrated the spectral features of nonstationary wind after two-directional decomposition, providing an input-validation baseline for the frequency-domain or evolutionary-spectral path. Data-driven simulation of full-scale downburst wind speeds [32] used stationary wavelet transforms and Hilbert transforms to decompose the nonstationary signal into mono-component signals, providing an input-construction method for the time-frequency path.

Table 1 . Impact matrix of thunderstorm wind-field elements on load and response computation stages

Wind-field element	Load time-history amplitude	Load frequency content	Spatial correlation structure	Response peak timing	Source keys
Time-varying mean	Directly controls envelope shape	Low-frequency modulation alters spectral shape	Affects weighting at each height through the profile	Determines the time of maximum mean response	[4] [21]
Moving path	Alters effective load duration	Introduces Doppler frequency shift	Asymmetric along the path direction	Ratio of translation velocity to structural width controls effective action time	[39] [28]
Duration	Limits degree of resonance accumulation	Determines lowest resolvable frequency	Not reported	Peak may occur during the ramp-up segment in short events	[45] [48]
Time-varying variance	Alters fluctuating-load peak values	Corresponds to evolutionary spectral shape change	Time-varying variance degrades coherence models	Variance peak and mean peak may be asynchronous	[4] [2]

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Wind-field element	Load time-history amplitude	Load frequency content	Spatial correlation structure	Response peak timing	Source keys
Turbulence intensity	Controls ratio of fluctuating component to mean	High turbulence intensity broadens the response spectrum	Affects spatial averaging coefficient	Resonant component relatively increases under high intensity	[45] [46]
Non-Gaussianity	Alters extreme-value distribution tails	Higher-order moments affect crossing rates	Non-Gaussian field requires joint-distribution description of spatial correlation	Peak-factor formulas require correction	[69] [109]
Spectra and evolutionary spectra	Frequency weighting determines modal contributions	Describes frequency migration of energy over time	Time-varying form of the cross-spectrum	Frequency band of energy concentration determines resonance intervals	[74] [53]
Time-varying coherence	Affects total load through resultant-force reduction	Time-varying coherence length alters generalized-force spectra	Core descriptive object	Coherence changes alter resultant-force peaks	[2] [97]
Spatial correlation	Area-averaging effect	Participates in frequency-domain computation through coherence functions	Core descriptive object	Correlation structure alters extreme-value combinations	[83] [102]
Nose-shaped profile	Alters load weighting at each height	Profile shape affects modal loads	Affects shear effects through profile gradient	Maximum-load height interacts with structural mode shapes	[6] [60]
Reference height	Alters the denominator of the pressure coefficient	Normalization affects dimensionless spectral shape	Indirect influence	Indirect influence	[118] [99]
Direction	Angle of attack determines aerodynamic force coefficients	Direction change introduces time-varying angle of attack	Cross-wind component coherence structure	Direction tracking affects peak combinations	[22] [118]
Radial position	Determines effective wind speed magnitude	Vortex-structure interaction frequency varies with position	Radial position alters correlation length	Source-local critical position controls loads	[30] [54]
Translation velocity	Alters effective action time	Introduces transient features	Alters coherence along the path	Translation velocity changes peak duration	[39] [27]
Ramp-up rate	Acceleration effect may alter aerodynamic forces	Rapid ramp-up broadens the effective frequency band	Asymmetric correlation during acceleration segment	Load overshoot when ramp-up rate is high	[24] [108]

Note: Cells are qualitative synthesis of source-local or method attributes and do not support cross-source quantitative derivation.

3 Wind-Speed-to-Pressure/Force Relationships

3.1 Wind-tunnel test route: pressure taps, force balances, aeroelastic and hybrid tests

3.1.1 Pressure taps and force balances

Pressure taps and force balances are the two basic routes for obtaining aerodynamic loads from wind-tunnel tests. Pressure taps retain the spatial distribution information of surface pressure and allow construction of resultant forces and moments over arbitrary integration surfaces during post-processing; force balances directly measure overall forces or segmental forces, with lower spatial resolution but stronger noise resistance.

Research on transient loads on buildings in microbursts and tornadoes recorded the ratio of force coefficients produced by translating microbursts on cube and tall-building models to quasi-static references [5]. When downburst-type accelerating flow reproduced in a multi-fan wind tunnel acted on a square-section model, the transient aerodynamic drag coefficient was lower than the stationary reference matched to the instantaneous wind speed; this difference was attributed to the acceleration-dependent wake state rather than an added-mass effect [24]. Research on time-varying extreme characteristics of a 5:1 rectangular cross-section under thunderstorm-type winds recorded load-statistic changes across different acceleration phases in a Tongji University multi-fan actively controlled wind tunnel [42]. Pressure distributions and net force coefficients of storm shelters obtained in thunderstorm wind-tunnel tests showed that downburst-induced roof uplift is significantly higher than the boundary-layer reference [36].

Research on transient wind loads on clusters of tall buildings [94] recorded the manifestation of inter-building interference effects under transient wind. A laboratory study of wind loads on low-rise buildings under a moving pulse jet simulator [101] compared pressure-coefficient differences obtained from different types of simulators; the moving jet and the stationary jet exhibit systematic differences in the temporal features of pressure peaks. Research on active flow modulation of large-scale turbulence effects on building pressures [104] demonstrated the mechanisms by which large-scale inflow fluctuations affect surface-pressure statistics.

3.1.2 Aeroelastic and hybrid tests

Aeroelastic tests allow structural motion to feed back into the aerodynamic forces, thereby capturing motion-dependent loads. Measurement of unsteady aerodynamic forces on a buffeting square cylinder using a hybrid aeroelastic force-balance method [130] revealed the sources of differences between base forces obtained from moving and fixed models; the hybrid method separates the aeroelastic force into motion-independent and motion-dependent components. Research on dynamic characteristics of aeroelastic transmission towers in synoptic wind and downburst-type outflows [57] recorded differences between damping ratio and frequency in downburst outflows versus those in conventional wind; the drag coefficient decreased by 20 to 48 percent under the downburst condition. Aeroelastic testing of transmission lines under multi-span downburst conditions [112] recorded the spatial distribution of conductor tension under multi-span coupling conditions on a scaled model. Research on the aeroelastic response of partially damaged and intact transmission towers under downburst-type outflows [25] showed that removal of local bracing members increased the root-mean-square acceleration by 86 percent in the along-wind direction.

3.1.3 Transient wind-generation devices

Differences among transient wind-generation devices determine the physical meaning and transferability of the load records they produce. A review of physical simulation of downburst wind fields [3] categorized existing devices into stand-alone types (gravity current, impinging jet, multi-fan) and combined types (active elements embedded in boundary-layer wind tunnels); stand-alone types are limited in scale ratio and controllable parameters, while combined types offer advantages in reproducing transient temporal features. Experiments on generating and characterizing nonstationary flows

in a small wind tunnel using a multi-blade device [43] provided an input-validation baseline for modal superposition; that device reproduced the basic features of the nose-shaped wind-speed profile in a 1.7 m wide, 1.35 m high test section by adjusting blade rotational-speed programs. Research on producing downburst outflows using multi-blade flow-control devices in conventional wind tunnels [100] recorded spectral-evolution differences produced by different blade rotational-speed programs. Research on aerodynamic forces on generic buildings in transient downburst-type winds [55] recorded the time-varying features of force coefficients across different building forms.

3.2 CFD route: RANS/URANS/LES and pressure integration

3.2.1 RANS and URANS

URANS analysis of a stationary impinging-jet model [12] provided the temporal-evolution characteristics of pressure along the radial direction, but its pressure-coefficient definition depends on the axial dynamic-pressure reference, which is not directly comparable with definitions used in boundary-layer wind tunnels. A numerical study of wind-pressure characteristics on yurt-shaped buildings under downburst winds [11] obtained different pressure distributions from six RANS turbulence models, effective only for a rigid model; the choice of turbulence model had a small effect on the positive-pressure zone of the windward face but a significant effect on the suction zone of the leeward face.

A numerical study of wind-load effects on buildings subjected to downbursts versus boundary-layer flow [18], in the form of a full-length thesis, provided numerical evidence for canonical geometries; RANS and hybrid RANS/LES results presented in different chapters require separate treatment. Research on downburst outflow models and wind pressures on tall buildings under different terrain conditions [67] demonstrated the modulating effect of terrain roughness on pressure distributions. A study of quasi-static parameters of wind turbines under downburst conditions [38] used a RANS wind field to drive a parked-state turbine model, obtaining the maximum tower-base bending moment at a source-specific radial position.

3.2.2 LES and pressure extraction

An experimental and numerical study of impinging-jet flow field and wind loads on cubic buildings in microburst-type winds [103] compared surface pressure distributions obtained from numerical and experimental methods. Large-eddy simulation of microburst wind flow around a building [15], starting from a mesoscale convective system, resolved the downburst flow field down to building scale through nested grids; that study treated only the two-dimensional cross-section case. A numerical study of downburst flow fields around low-rise buildings using large-eddy simulation [19] compared peak force coefficients for a cube and a portal-frame building within same-scale impinging jets.

A verification study on the prospects of LES for predicting wind loads and response of tall buildings [23] established a four-stage verification chain from inflow to response on the CAARC benchmark, separately evaluating inflow-turbulence reproduction, pressure statistics, base-force spectra, and dynamic response. A verification study of standard CFD setups for wind-load assessment of tall buildings [91] provided grid and subgrid-model sensitivity references for the CAARC benchmark.

3.2.3 Grid, time step, and turbulence-model sensitivity

Within the CFD route, the choices of grid resolution, time step, and turbulence model introduce methodological sensitivity, and these three are coupled: grid refinement requires the time step to be simultaneously reduced to maintain the Courant-number constraint, and the filter scale of the subgrid model is directly linked to the grid size. A CFD study of a large semi-open canopy roof under nonstationary wind [14] addressed unsteady aerodynamic effects; load-exceedance effects during the rapid ramp-up phase are applicable only to that specific stadium-canopy geometry. Numerical pressure analysis of a rectangular cross-section in a moving thunderstorm downburst wind field [39] used LES

with a dynamic-mesh method to simulate a moving nozzle; the grid strategy and time-step selection determine the resolution precision of vortex structures. Within the RANS and URANS frameworks, the turbulence model choice determines the closure of the Reynolds-stress tensor: standard two-equation models have limited predictive ability in regions of strong adverse pressure gradients and large separation, while Reynolds-stress transport models, though theoretically more complete, rely on experimental data from stationary-flow conditions for parameter calibration. The subgrid-model sensitivity in LES concentrates on the method for generating inflow turbulence and the grid resolution capability near the wall; when the inflow conditions use a synthetic-turbulence method, the influence of synthesis parameters on near-body pressure statistics may be of the same order as the influence of grid resolution [23] [91]. Grid-convergence verification faces additional difficulty in transient wind fields: because the input itself is nonstationary, traditional grid-independence checks must be performed in a statistical sense rather than in a single-time-history sense, requiring multiple simulations to obtain converged statistics, which further increases computational cost.

3.3 Theoretical and analytical route: quasi-steady, analytical downburst profiles, generalized forces, and equivalent static forces

3.3.1 Quasi-steady assumption and analytical profiles

The quasi-steady assumption represents nonstationary loads as the product of the instantaneous wind speed and the stationary aerodynamic force coefficient. A numerical study of wind loads on buildings induced by downbursts [137] computed surface pressures on low-rise buildings within the quasi-steady framework; the quasi-steady assumption approximately holds when the rate of wind-speed change is lower than the flow-field development time scale. Numerical and analytical simulation of downburst wind loads [134] compared base-shear time histories given by different wind-field models; the parametric flexibility of analytical profile models leads to different load estimates for the same building under different analytical models.

Research on surface pressures and wind loads on prismatic bodies in transient gust-front flows [80] recorded the rapid reorganization of the pressure distribution as the gust front passes, with the deviation of the quasi-steady assumption being most significant during the passage of the leading-edge vortex. Experimental study of pressure distributions on prismatic bodies in experimentally simulated downbursts [140] provided supplementary surface-pressure evidence for low-rise buildings from an experimental perspective.

3.3.2 Generalized-force and equivalent-static-force construction

The construction of generalized forces depends on mode shapes, and modal-truncation criteria require re-examination under nonstationary input. Research on transient aerodynamic forces on a square cross-section under downburst-type accelerating flow in a multi-fan wind tunnel [110] demonstrated the evolution of the generalized-force spectrum across acceleration phases. A review and full-scale verification of analytical wind-load computation methods for downbursts [121] noted that when different analytical models are adopted for load computation, the frequency content of the resulting generalized forces changes.

The construction of equivalent static loads depends on the choice of the target response quantity; when the target changes, the equivalent distribution changes accordingly. The computational framework for the generalized gust-front factor [58] and along-wind response analysis of tall buildings under transient nonstationary wind [59] respectively treated nonstationary equivalent static demands from framework and case perspectives. The gust-front factor framework [131] provided a new organizational scheme for nonstationary wind-load effects, linking equivalent static loads to transient peak factors.

3.4 Comparison of validation strength, transferability, and uncertainty across the three route families

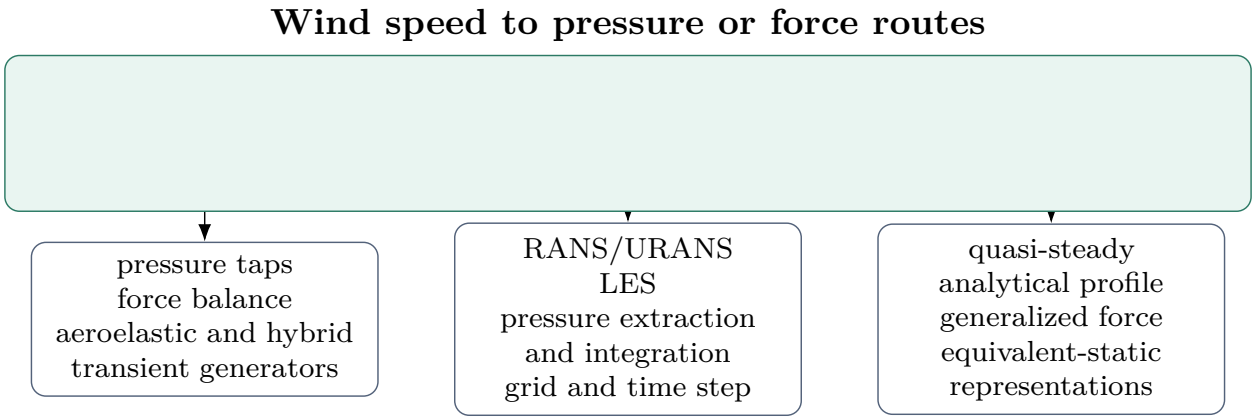


Figure 2 Classification tree for wind-speed-to-pressure-or-force routes. Original synthesis based on exact146 accepted-only evidence; does not redraw source numerical curves and does not represent a PETW computation result. The top level divides into wind-tunnel route (pressure taps, force balance, aeroelastic or hybrid, transient generation device), CFD route (RANS, URANS, LES, pressure extraction and integration), and theoretical or analytical route (quasi-steady, analytical profile, generalized force, equivalent static force). Each sub-node is annotated with the source-specific information-retention level and typical validation method.

Table 2 . Comparison of three wind-speed-to-pressure-or-force route families

Comparison axis	Wind-tunnel route	CFD route	Theoretical and analytical route
Information retention	Pressure taps retain spatial distribution; force balance retains overall force	Full-field pressure and velocity extractable, limited by grid resolution	Typically retains only overall force or generalized force
Modeling assumptions	Physical similarity laws determine the validity range	Turbulence model and boundary conditions constitute assumptions	Quasi-steady or analytical profile is the core assumption
Validation basis	In-laboratory repeatability and cross-facility comparison	Comparison with experiment or field measurements	Comparison with experimental or numerical results
Cost	Model fabrication, apparatus operation, and measurement equipment	Grid generation, computation time, and storage	Lowest computational cost
Transferability	Limited by apparatus and scale ratio	Limited by grid and turbulence model choices	Parameterized profiles are adjustable but require validation
Uncertainty sources	Scale effects, Reynolds-number mismatch, apparatus repeatability	Grid, time step, turbulence model, inflow conditions	Assumption applicability, parameter selection
Representative sources	[5] [24] [36] [43] [55]	[12] [15] [19] [23] [91]	[137] [134] [121] [58] [59]

Note: Cells are qualitative synthesis of source-local or method attributes and do not support cross-source quantitative derivation.

Mapping from building-cluster pressure distributions to concentrated nodal forces involves the choice of area-integration scheme. A numerical study of impinging-jet models acting on different array configurations [71] showed that mutual shielding among buildings alters the surface-pressure distribution of individual buildings, so the nodal forces obtained by integration depend on building spacing and arrangement angle. Surface-pressure-distribution data obtained on a cube in a simulated

moving downburst [72] showed that moving and stationary downbursts differ in the temporal characteristics of pressure-peak time series. Analysis of aerodynamic force characteristics of trains in a stationary thunderstorm downburst wind field [81] demonstrated the time-varying pattern of forces on non-building objects in downbursts.

A numerical study of wind-pressure characteristics on yurt-shaped buildings under downburst winds further showed [11] that different RANS turbulence models have different degrees of influence on the windward and leeward faces. Aerodynamic characteristics of tall buildings in stationary thunderstorm outflow-type flow fields [116] recorded the distribution of pressure coefficients with height under a nose-shaped wind-speed profile. An experimental study of flow field and wind loads on gable-roof buildings in microburst-type winds [117] demonstrated the effect of roof slope on peak suction values. Experimental wind-load evaluation of elevated buildings in boundary-layer and downburst winds [62] compared the pressure differences of the same building model in two types of wind fields.

An experimental study of wind-induced loads on tall buildings under downburst outflow conditions [29] measured the base-shear dynamic amplification factor for the CAARC model under three outflow environments in a multi-fan wind tunnel; the unified base-shear dynamic amplification factor produced by the translating outflow was 2.18. Research on wind-pressure distributions and wind-field response of a large-span hyperbolic-paraboloid shell roof under downburst conditions [119] recorded the influence of sloped terrain on large-span roof pressures. Research on pressures on a large-span hyperbolic spherical-shell roof [77] further showed that terrain and radial position alter the pressure zoning between windward and leeward regions.

An analytical load model for bridge decks under downbursts [50] mapped radial-velocity profiles to span-wise distributions, but the one-dimensional wind-field and quasi-steady aerodynamic-force assumptions limit its applicability range. A hybrid CFD and structural-dynamics path [90] directly transferred the transient flow-field output to the bridge model. Nonstationary bridge-deck aerodynamic forces under transient wind recorded vortex memory and path dependence [84]. A non-negative matrix factorization method for multi-point wind fields [141] extracted low-rank spatial structure from the observation matrix.

The six RANS turbulence-model differences used in a numerical study of yurt-shaped buildings under downburst wind [11], a field study of wind and structural response of lighting poles in downbursts [86], a PIV analysis of experimentally simulated downburst-type wall jets in a large-scale experimental facility [75], and research on the significance of convective gusts [48] respectively provided supplementary evidence for the wind-speed-to-pressure route from the perspectives of RANS model differences, full-scale records, laboratory flow-field features, and meteorology.

Fine analysis of thunderstorm outflow characteristics [129] discussed the association between base-force time histories and inflow feature parameters. Research on peak-factor and gust-factor estimation for nonstationary wind speeds [132] noted that the definition of the peak factor in a nonstationary process itself depends on the selected time window and mean definition; this dependence affects the normalization of force coefficients. Numerical and analytical simulation of downburst wind loads [134] compared base-shear time histories given by different wind-field models. Experimental and numerical evaluation of transmission-tower tip response [136] demonstrated the allocation of base forces in the longitudinal and transverse directions from the tower-conductor system perspective.

The normalization baselines for pressure coefficients and force coefficients differ among sources (axial dynamic pressure, peak-height dynamic pressure, 10 m mean dynamic pressure, etc.); therefore, force-coefficient values from different sources cannot be directly juxtaposed. Quantitative transfer among building-type sources [94] [14] [18], bridge-type sources [50] [95], and wind-turbine-type sources [38] [98] requires prior alignment of normalization baselines.

The response of wind turbines is jointly determined by rotor operating state, control strategy, and tower dynamic characteristics. Dynamic analysis of a horizontal-axis wind turbine [126] reported blade-root bending moment and tower-base shear time histories under parked conditions. A fluid-structure-interaction scaled model [52] recorded the bidirectional interaction between aerodynamics and structural motion. These results cannot be compared independently of whether the rotor is

spinning, whether pitch control is active, whether the wind field includes directional changes, and the aerodynamic-model order.

Research in life-cycle cost assessment regarding vertical structures under nonstationary wind [127] compared the differences in generalized-force construction between downburst and tornado loads. A hybrid method based on empirical mode decomposition and spectral representation for data-driven simulation of multivariate nonstationary wind [139] showed that different time-varying-mean extraction methods produce different fluctuating components, which in turn affect generalized-force construction. A review of nonstationary wind modeling [143] discussed the influence of nonstationary-process characterization on equivalent static demands.

Research on roof aerodynamic forces on buildings of different heights in transient wind [145] recorded the interaction effect between roof height and transient wind profile. A study of wind-pressure characteristics on yurt-shaped buildings under downburst wind [11] and nonstationary vortex-shedding effects in non-synoptic wind response modeling of tall buildings [20] respectively demonstrated the behavior of unconventional buildings and vortex-shedding loads under downburst conditions. Research on a nonstationary turbulent wind-speed model and its influence on wind-induced response of tall buildings [41] compared the response differences produced by four wind-speed models on the same structure.

Research on dynamic effects of downburst wind on longitudinal forces in transmission towers [7] discretized the conductor across each 500 m span into 90 cable elements in a six-span continuous model. Full-scale downburst wind loads measured on a transmission line in northern Germany [146] provided one of the few full-scale records available. Experimental and numerical evaluation of transmission-tower tip response [136] and research correlating experimental simulation of downburst outflows with dynamic characteristics of transmission towers [133] measured the frequency-domain features of tower-body dynamic response in the wind tunnel.

Research on conductor response under downburst wind [61] analyzed the dynamic tension changes of conductors at different positions along the moving downburst path. Comparison of input and structural characteristics under three outflow configurations in downburst outflow [125] provided measured-versus-numerical comparisons.

Parameterization of downburst loads in industry [98] can be used to identify input-variable categories, but its clause context and applicable object do not constitute cross-industry response criteria. Response-history analysis and statistical linearization for inelastic response of base-isolated tall buildings under nonstationary wind [78] provided probabilistic descriptions of response quantities on a nonlinear path.

4 Pressure-to-Structural-Degrees-of-Freedom Interface and Equivalent-Load Construction

4.1 Distributed pressure to nodal loads

The mapping from distributed pressure to concentrated nodal forces involves the choice of area-integration scheme. The spatial density of tap placement, the choice of reference dynamic pressure, and the definition of the time window together determine the physical meaning of the pressure coefficient. The layout of wind-tunnel pressure taps determines the upper limit of area-integration accuracy, while in the CFD route, pressure on grid faces can in principle provide higher-resolution spatial distributions. Research in life-cycle cost assessment regarding vertical structures under nonstationary wind [127] showed, in the pressure-to-generalized-force mapping, that the degree of spatial matching between mode shapes and the pressure distribution determines the signal-to-noise ratio of the generalized force.

Research on transient aerodynamic forces on a square cross-section under downburst-type accelerating flow in a multi-fan wind tunnel [110] demonstrated the evolution of the generalized-force

spectrum across acceleration phases; the effectiveness of the pressure area-integration scheme at different time stages depends on the ratio of tap spacing to pressure correlation length. A review and full-scale verification of analytical wind-load computation methods for downbursts [121] noted that different analytical models produce different frequency contents in the generalized forces.

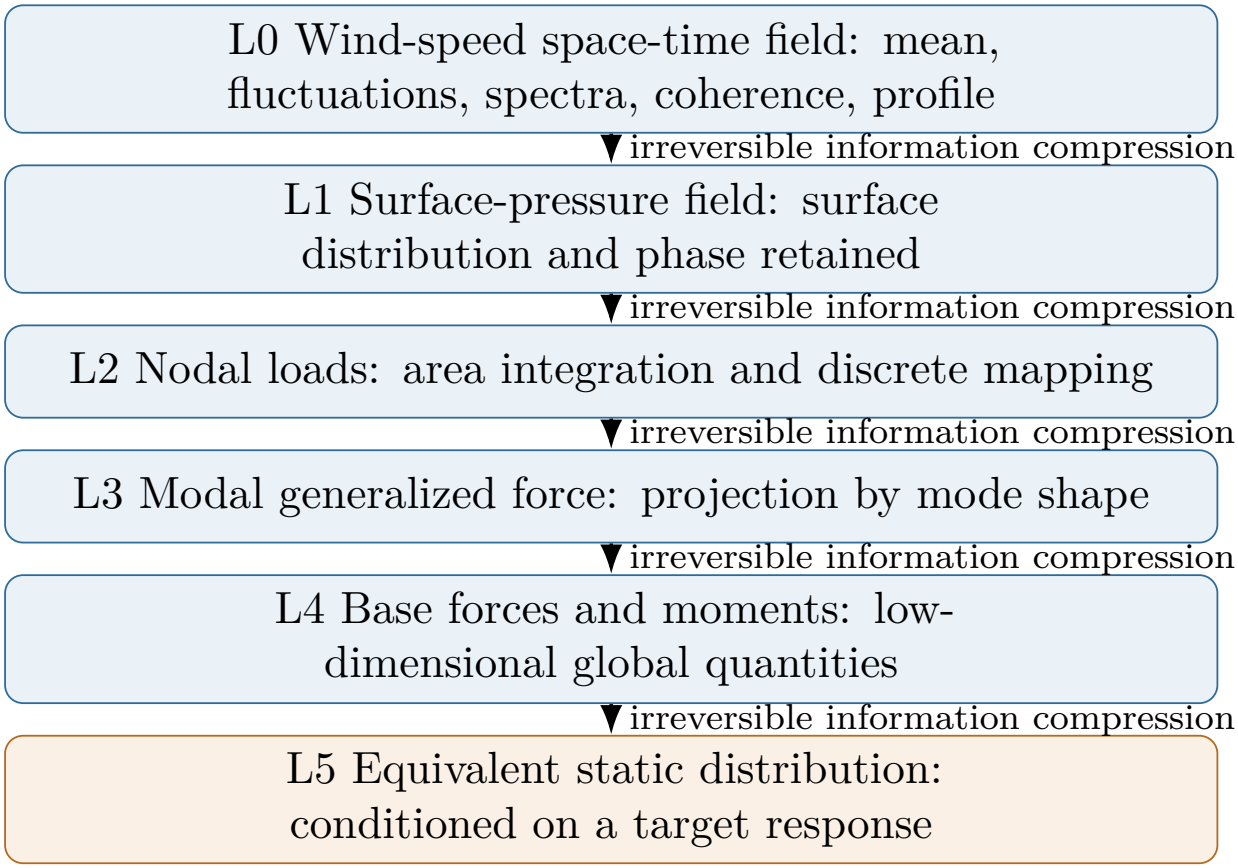


Figure 3 Schematic of input-object hierarchy and information loss. Original synthesis based on exact146 accepted-only evidence; does not redraw source numerical curves and does not represent a PETW computation result. From top to bottom: wind-speed field (preserves full spatiotemporal information), pressure field (preserves surface spatial distribution), nodal forces (discrete forces after area integration), generalized forces (scalar time histories after modal projection), base forces (all spatial information compressed into low-dimensional vectors), and equivalent static forces (static force distributions conditioned on specific response quantities). Each step of information compression is irreversible; downstream levels cannot recover the spatial or frequency information already lost at upstream levels.

4.2 Nodal loads to modal generalized forces

The construction of generalized forces depends on mode shapes, and modal-truncation criteria require re-examination under nonstationary input. When input energy is concentrated in a specific frequency band during a certain time segment, only modes with significant generalized-force components in that frequency band make substantial contributions to the response; however, because the energy-concentration band itself shifts over time, a static modal-truncation order may be insufficient at some instants and redundant at others.

Research in life-cycle cost assessment on the differences in generalized forces under nonstationary wind loads [127] demonstrated the different modal contribution allocations under downburst and tornado inputs. A review of nonstationary wind modeling [143] discussed the influence of nonstationary-process characterization on modal-truncation requirements. Research on peak-factor and gust-factor estimation for nonstationary wind speeds [132] noted that the peak-factor definition in a nonstationary process itself depends on the selected time window and mean definition, affecting the determination of generalized-force peaks. A hybrid method based on empirical mode decomposition and spectral

representation [139] demonstrated the influence of different time-varying-mean extraction methods on the fluctuating component and thereby on generalized-force construction. Multi-fan wind-tunnel pressure measurements [102] provided reference data for multi-point generalized-force construction.

4.3 Base forces, moments, and spatial information loss

Base forces occupy a specific interface position as intermediate response quantities. Base shear and overturning moment are scalar or low-dimensional vector results of spatial integration of pressures; in this process, the spatial-distribution information of the pressure is lost. Fine analysis of thunderstorm outflow characteristics [129] focused on the association between base-force time histories and inflow feature parameters. Numerical and analytical simulation of downburst wind loads [134] compared base-shear time histories given by different wind-field models. Experimental and numerical evaluation of transmission-tower tip response [136] demonstrated the allocation of base forces in the longitudinal and transverse directions from the tower-conductor system perspective.

4.4 Response-target dependence of ESWL/GFF/G-GFF representations

Equivalent static wind loads, gust-front factors, and generalized gust-front-factor representations are all conditioned on specific target responses. When the target shifts from roof-top displacement to base moment, local member forces, or cladding pressure, the equivalent distribution and peak-combination rules change accordingly; therefore, such representations cannot be understood as unique loads independent of the response target. Related frameworks can be used to illustrate the mathematical link between input compression and target response [131] [58] [59], but source-local constructions cannot be converted into PETW design coefficients or cross-archetype universal quantities.

4.4.1 Time synchronization and dimensional consistency

Time synchronization and dimensional consistency determine whether the load interface can be received losslessly by the structural solver. If pressure taps, force balances, and displacement sensors use different sampling frequencies, they must be mapped to a common time axis after anti-aliasing filtering; simple linear interpolation, while maintaining temporal correspondence, may attenuate local peaks in fast ramp-up segments. Spatial mapping must also maintain resultant-force and resultant-moment balance; otherwise, the nodal loads are not equivalent to the original pressure field in the overall sense. For studies using moving wind fields or moving jets, the transformation between load and structural coordinate systems varies with time, and radial, tangential, and vertical components cannot be combined at a fixed angle of attack. The core issue of dimensional consistency lies in the definition of the reference dynamic pressure: different sources may use peak wind speed, time-averaged wind speed, or jet exit velocity as the normalization baseline, producing force coefficients whose numerical values are not directly comparable. When pressure coefficients and velocity time histories come from different experiments or different simulations, inconsistency in the normalization baseline may introduce systematic bias. Furthermore, the matching between the geometric scale ratio and the temporal scale ratio of the wind-tunnel model is an important element of dimensional consistency: if the geometric scale ratio is 1:200 but the temporal scale ratio is not adjusted synchronously according to Froude number or reduced-frequency requirements, the load time history at prototype scale will be distorted on the frequency axis.

Related studies respectively showed, from multi-point nonstationary simulation [70], pressure measurement [2], and structural generalized-force construction [110] perspectives, that the choices of time window, reference velocity, and spatial coherence model simultaneously alter load amplitude and phase relationships. Therefore, the interface check must verify not only file format but also units, sign conventions, coordinates, sampling, reference dynamic pressure, area weights, resultant-force balance, and phase consistency.

5 Linear Response Computation Methods

5.1 Time-domain direct integration and modal superposition

The time-domain direct-integration method is the most straightforward solution path in transient response analysis. This method iterates the equation of motion at each time step using numerical schemes (e.g., Newmark-beta or Wilson-theta), requiring no a priori assumptions about the statistical properties of the input process and thus being applicable in principle to arbitrary nonstationary input. However, the computational cost of the time-domain method is proportional to the product of time-step count and degrees of freedom, and for probabilistic analyses requiring large numbers of sample time histories to obtain statistically stable results, computational cost may become a limiting factor. Moreover, although the introduction of numerical damping can suppress high-frequency numerical oscillation, excessive numerical damping artificially attenuates high-frequency response components; in the fast ramp-up segments of transient input, particular attention is needed to the trade-off between time step and numerical damping.

The time-domain direct-integration method, given step-by-step load time histories, solves the equation of motion by numerical iteration, with step-size selection and numerical-damping settings directly affecting solution accuracy. Boundary-layer wind-tunnel experimental simulation of thunderstorm profiles used passive grids to generate quasi-static nose-shaped wind-speed profiles [6]; such stationary devices produce load time histories usable as time-domain integration input but lack nonstationary temporal-modulation features. Review literature on downburst wind-field characteristics and structural wind effects organized the application of time-domain computation paths across different structural archetypes [28].

Data-driven simulation of full-scale downburst wind speeds [32] used stationary wavelet transforms and Hilbert transforms to decompose the nonstationary signal into mono-component signals, extracting instantaneous amplitude and instantaneous frequency and then generating sample time histories satisfying target statistical properties. A computer model for digitally simulating the transient dynamics of a tall building under Andrews Air Force Base thunderstorm conditions [63] solved the roof-top displacement and acceleration response in the time domain. Research on the response of linear systems under nonstationary random filtered input [8] provided a theoretical basis for the applicability of the linear superposition principle under filtered excitation.

5.2 Frequency domain, evolutionary spectra, and background-resonance decomposition

Frequency-domain and evolutionary-spectral methods describe the nonstationary input using evolutionary power spectral density and obtain the evolutionary spectrum of the response through a time-varying frequency-response function. The core premise of this family of methods is that the input process be expressible in a uniform-modulation or slowly-varying-modulation form; when the modulation rate is of the same order as the structural fundamental period, the slowly-varying assumption degrades. The time-frequency spectral representation model [44] provided a theoretical framework for the simulation of nonstationary processes, with the core requirement being the separation of instantaneous frequency and amplitude-modulation function.

An evolutionary PSD model [53] was constructed under conditions consistent with full-scale records, with parameters confined to specific measurement stations and event samples. Research on simulating fully nonstationary random processes using generalized harmonic wavelets [31] demonstrated the evolutionary-spectral convergence accuracy at 1,000 samples for nonstationary-process sample generation. A stochastic harmonic-function representation of nonstationary random processes [51] provided an alternative path for generating nonstationary realizations from the frequency-domain perspective. Research on the significance of convective gusts [48] emphasized the fundamental difference between the frequency content of transient loads and the stationarity assumption.

Fast convolutional-integral nonstationary response analysis of non-proportionally damped structures [13] provided benchmark comparisons of computational efficiency under both linear and nonlinear conditions; the presence of non-proportional damping means that equations in modal space are no longer strictly decoupled. Research on the influence of uncertain models of thunderstorm wind speed and uncertainty propagation on maximum dynamic response [26] demonstrated the response intervals of linear-elastic models under epistemic uncertainty on two slender structures.

5.3 Wavelet, S-transform, Hilbert route, and time-frequency response identification

Time-frequency decomposition methods occupy a different position in response solution compared with frequency-domain methods. Transient aeroelastic research [64] advanced the analysis of structural response under thunderstorm outflows into the time-frequency domain, where the instantaneous frequency of structural response can be tracked. The evolutionary PSD model for thunderstorm outflows [74] and PIV analysis of experimentally simulated downburst-type wall jets in a large-scale experimental facility [75] respectively provided reference comparisons of time-frequency features from field and laboratory dimensions.

Wavelet and transform singular-value-decomposition system-identification methods for nonstationary response [37] mapped the nonstationary response to the time-frequency domain using the continuous Morlet wavelet transform, then used transform singular-value decomposition to automatically separate closely spaced or overlapping modal components. A nonstationary frequency-domain system-identification method [88] used time-frequency representations to extract time-varying modal parameters. A time-frequency simulation study [40] fed a measured derecho downburst record and a parameterized extreme-wind model input into the same 5 MW turbine finite-element model, comparing displacement and acceleration.

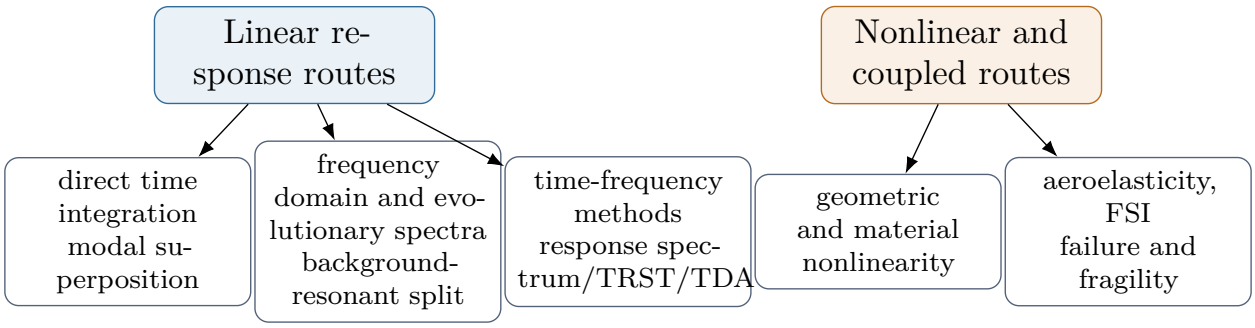


Figure 4 Classification tree for structural response computation methods. Original synthesis based on exact146 accepted-only evidence; does not redraw source numerical curves and does not represent a PETW computation result. The top level divides into linear methods and nonlinear methods. Under linear methods: time-domain direct integration, frequency-domain or evolutionary-spectral, time-frequency decomposition, and response-spectrum or TRST/TDA, four routes. Under nonlinear methods: geometric nonlinearity, material nonlinearity, aerodynamic feedback and aeroelasticity, progressive-collapse paths, and reduced-order methods. Each branch is annotated with core assumptions and representative source keys.

5.4 TRST/TDA/response-spectrum path and applicability boundary

The transient response-spectrum method provides a simplified path without performing a full time-domain analysis. An experimental study of near-ground flow dynamics of downburst-type impinging jets [106] provided input descriptions for the response-spectrum method. Research on the importance of thunderstorms in wind engineering and the needs for next-generation wind tunnels [122] discussed the applicability conditions and limitations of the transient response-spectrum method. A comparison of the response-spectrum technique and time-domain analysis for thunderstorm outflows [76] compared peak-response estimation differences between the two paths.

The premise of background-and-resonance decomposition is that the input spectral shape remains sufficiently stable over the time segment of interest. Under transient-input conditions, the definition of the resonant component itself becomes time-varying, and the interface between background and resonant components shifts with the modulation phase. Frequency-domain analysis of transmission towers constructs equivalent demands from background and resonant responses [35].

5.5 Comparability and error sources among linear methods

The assumption boundaries of linear methods can be summarized as three premises: small deformation, classical damping, and decoupling of load and response. Dynamic response analysis of conductors during transmission-tower collapse [142] revealed that once the tower body enters the nonlinear failure stage, conductor response departs from the applicable range of linear superposition. Research on a nonstationary turbulent wind-speed model and its influence on wind-induced response of tall buildings under nonstationary wind [41] compared the response differences produced by four wind-speed models on the same structure; matching the mean and turbulence intensity does not guarantee consistent response.

In transient response studies, TRST and TDA are commonly used to refer to response-spectrum or time-domain analysis paths, but different sources are inconsistent in their reporting of full abbreviation definitions, input construction, and peak-combination rules. When sources do not report damping corrections, modal-combination rules, or spectrum-construction details, the numerical results of TRST and TDA can only be taken as source-local results [76] [13] [8].

Table 3 . Comparison of structural response computation methods

Method category	Input requirements	Core assumptions	Output level	Validation method	Computational cost	Representative sources
Time-domain direct integration	Step-by-step load time history	Small deformation, classical damping	Complete response time history	Time-step and grid convergence	High	[63] [32]
Frequency-domain or evolutionary-spectral	Evolutionary power spectral density	Slowly-varying modulation, Gaussian	Response second-order moments	Comparison with time domain	Medium	[44] [31] [53]
Time-frequency decomposition	Time-frequency joint representation	Decomposition reversibility	Time-varying modal parameters	Component energy conservation	Medium-high	[64] [37] [88]
Response spectrum or TRST/TDA	Spectralized input	Peak-combination rules	Peak estimate	Comparison with time domain	Low	[76] [106] [122]
Geometric nonlinearity	Step-by-step load time history	Large deformation, updated configuration	Nonlinear response path	Incremental convergence, path comparison	High	[9] [123]
Aeroelastic and FSI	Coupled load and motion	Motion-dependent aerodynamic forces	Coupled response	Independent aerodynamic-force validation	Very high	[64] [92] [25]
Reduced-order method	Simplified model parameters	Projection accuracy	Approximate response	Comparison with full-order model	Medium-low	[107] [13]

Note: Cells are qualitative synthesis of source-local or method attributes and do not support cross-source quantitative derivation.

6 Nonlinearity, Aerodynamic Feedback, and Failure Mechanisms

6.1 Geometric and material nonlinearity

Nonlinear response analysis holds special importance under transient wind action. When loads increase rapidly, the structure may undergo a transition from elastic to plastic or from stable to unstable within a short time, and the linear superposition principle fails in this process. Nonlinear analysis requires updating the stiffness matrix at each time step or increment, with computational cost significantly higher than linear analysis, but the information it provides is critical for understanding structural behavior under extreme loads. Nonlinear problems can be classified by physical source into geometric nonlinearity (stiffness-matrix change due to large deformation), material nonlinearity (nonlinear stress-strain relationships), and contact nonlinearity (boundary-condition changes with deformation); in tower-conductor systems, these three types of nonlinearity often appear simultaneously.

Geometric nonlinearity manifests in large-displacement cable-strut systems and flexible structures as response features in which stiffness varies continuously with configuration. Research on time-varying structural reliability under nonstationary non-Gaussian processes [9] proposed analytical methods for crossing rate and first-passage probability; that method is applicable under conditions of weakened mean-crossing rate and Poisson assumptions for non-Gaussian response processes caused by geometric nonlinearity. An investigation of downburst-related damage in Brazilian buildings [17] recorded the failure patterns of low-rise building roof structures under extreme wind speeds, of which roof-panel connection failure was the most common damage mode; that observational result is confined to the specific construction conditions of the investigated buildings.

Material nonlinearity and damage accumulation become the dominant mechanism once the structural response has entered the plastic-development stage. A computationally efficient stochastic method for the vulnerability of vertical structures under downburst wind [79] introduced stochastic nonstationary excitation within a Monte Carlo simulation framework; its vulnerability curve is conditioned on peak wind speed. A multi-hazard analysis framework for earthquake-damaged tall buildings subjected to thunderstorm downbursts [113] used the residual-damage state as the initial condition for subsequent wind-load response; in that analysis, material nonlinearity was expressed using a fiber-section model.

6.2 Motion-dependent aerodynamic forces, aeroelasticity, and FSI

The appearance of aerodynamic feedback and motion-dependent loads signals the failure of the premise that load and response are decoupled. Research on collapse mechanisms and failure criteria of super-large cooling towers subjected to downbursts [73] recorded the influence of load-application paths on internal-force distributions, but that study adopted a quasi-static load-application method rather than dynamic coupling. Discovery of time-varying aerodynamic forces from bridge prototypes based on sparse identification of nonlinear dynamics [92] demonstrated the technical path for identifying motion-dependent aerodynamic forces from response data. Fast convolutional-integral nonstationary response analysis of non-proportionally damped structures [13] provided computational-efficiency benchmarks under the nonlinear case.

Estimation of nonstationary cross-wind response of tall buildings under nonlinear aeroelastic effects [87] calibrated the degree of influence of vortex-induced-force model parameters on response through parameter sweeps. Research on vulnerability surfaces of suspension bridges under the dual hazard of earthquakes and microbursts [10] showed that the choice of damage-state variable determines the shape and conditional meaning of the vulnerability surface. The identification and modeling of motion-dependent loads face two difficulties: first, the nonstationarity of the transient wind input itself causes the system parameters to vary with time, so the classical frequency-domain definitions of aerodynamic derivatives or aerodynamic admittance must be extended to a time-varying framework; second, when structural displacements are large, geometric nonlinearity and aerodynamic nonlinearity

act simultaneously, and the response increment cannot be attributed to a single mechanism. When the fundamental frequency of the structure approaches the vortex-shedding frequency range, lock-in causes the frequency content of the aerodynamic force to be entrained by the structural response, and a load model independent of the response will underestimate the amplitude. For flexible structures such as transmission lines and suspension bridges, the change in projected area and drag coefficient caused by large-amplitude swings constitutes another layer of motion-dependent effect; this effect is especially pronounced under the strong vertical component of thunderstorm outflows.

6.3 Failure, collapse, vulnerability, and multi-hazard state dependence

Progressive-collapse and failure-path analysis reveals the path-dependent relationship between initial failure location and subsequent propagation. Research on the aeroelastic response of partially damaged and intact transmission towers under downburst-type outflows [25] showed that removal of local bracing members significantly alters the dynamic characteristics. Analytical prediction of quasi-static response of transmission lines subjected to moving downbursts [27] employed both nonlinear models and linear approximations. Progressive-collapse analysis of truss transmission tower-line systems under downburst loads [123] tracked the complete path from initial member buckling to system instability.

Analysis of wind-induced vibration and collapse failure of prestressed string structures under thunderstorm downbursts [120] showed that the relationship between the proportion of cable-force loss and the final structural response is nonlinear. A reduced-order wavelet Galerkin solution for coupled nonlinear stochastic response of slender buildings under transient wind [107] established a specific trade-off between computational efficiency and accuracy.

6.4 Verification of mechanism evidence and non-transferable boundaries

The evidence strength of nonlinear analysis depends on whether the mechanism has been independently constrained. Material constitutive relationships can be calibrated through component or material tests; geometric nonlinearity can be checked through grid and increment convergence; instability and collapse paths require sensitivity analysis of initial imperfections, connection stiffness, contact treatment, and failure-deletion rules. The successful completion of a single nonlinear solver run does not by itself establish the uniqueness of the obtained failure mechanism. Tower-conductor system and shell cases show that changes in initial imperfection location and load-movement path may alter the first-failing member and subsequent propagation sequence [123] [73]. Reliability or vulnerability analysis further superimposes model uncertainty with stochastic input, but the resulting curves remain confined to the adopted limit-state definition and the sample-generation model [9] [79].

Transient aeroelastic research [64] advanced structural response under thunderstorm outflow conditions into the time-frequency domain. Research on longitudinal forces in transmission towers caused by asymmetric downburst conductor loads [30] determined, through parameter sweeps, the critical angle of attack and radial distance conditions that maximize longitudinal force. The crossing-rate method for time-varying structural reliability under nonstationary non-Gaussian processes [9] and stochastic analysis of vertical-structure vulnerability under downburst wind [79] respectively treated exceedance probability under nonlinear conditions from analytical and numerical perspectives.

Research on dynamic characteristics of aeroelastic transmission towers in synoptic wind and downburst-type outflows [57] recorded changes in damping ratio and frequency in downburst outflows. Research on conductor response under downburst wind [61] and analysis of quasi-static buffeting response of transmission lines under moving downbursts [85] respectively demonstrated conductor dynamic tension and background fluctuating response. Aeroelastic testing of transmission lines under multi-span downburst conditions [112] recorded the spatial distribution of conductor tension under multi-span coupling conditions on a scaled model.

The hybrid aeroelastic force-balance method [130] revealed the sources of differences between base forces obtained from moving and fixed models. Three-dimensional dynamic analysis of nonstationary

downburst response of a transmission tower with separation of background and resonant components [34] demonstrated the application of the evolutionary-spectral-based method on a tower body. Research on wind-induced vibration and control of string structures under thunderstorm downbursts [135] recorded the peak-factor variation in dome structures.

Probabilistic study of nonstationary extreme-value response of vertical structures under downburst wind [47] assessed the uncertainty in extreme-value distribution parameters under limited event numbers. Dynamic response analysis of conductors during transmission-tower collapse [142] showed that after the tower body enters the collapse stage, the time-history characteristics of conductor tension undergo a fundamental change. Dynamic response analysis of transmission towers under downburst wind-wave loads [66] considered the coupled input of wind and waves in an offshore environment. Nonlinear numerical analysis of wind-pressure characteristics on yurt-shaped buildings under downburst wind [11] is effective only for a rigid model.

Comparison of transient response of bridges under downbursts and typhoons [95] showed that the two types of input have different time-frequency structures. A surrogate model study for long-span bridges under downburst conditions [68] reduced the cost of repeated solutions through parametric optimization. Numerical pressure analysis of a rectangular cross-section in a moving downburst wind field [39] provided supplementary evidence of vortex-structure interaction under moving conditions.

Inelastic response of base-isolated tall buildings under nonstationary wind [78] provided probabilistic descriptions of response quantities on a nonlinear path. Nonstationary vortex-shedding effects [20] showed limited contribution to cross-wind response under downburst conditions. Comparison of numerical and analytical simulation of downburst wind loads [134] provided comparisons of base shear under different wind-field models. Research on the behavior of self-supporting transmission towers under stationary downburst loads [124] recorded the critical load level at which the tower body enters the nonlinear range under different downburst sizes.

Research correlating experimental simulation of downburst outflows with dynamic characteristics of self-supporting transmission towers [133] measured the frequency-domain features of tower-body dynamic response in the wind tunnel. Critical load cases for lattice transmission-line structures under downburst action [49] discussed the controlling relationship of different downburst positions on structural demands. A finite-element optimization coupling technique for determining critical downburst parameters [54] used an optimization algorithm to locate the microburst position parameters that maximize tower-body response. Full-scale downburst wind loads measured on a transmission line in northern Germany [146] provided a full-scale record.

Research on response and first-passage probability of a linear-elastic SDOF system under nonstationary random excitation based on the S-transform [144] provided a computational path for first-passage probability in the time-frequency domain. Dynamic analysis of a horizontal-axis wind turbine [126] reported blade-root bending moment and tower-base shear time histories under parked conditions. Discovery of time-varying aerodynamic forces from bridge prototypes based on sparse identification of nonlinear dynamics [92] demonstrated the technical path for identifying motion-dependent aerodynamic forces from response data.

Nonstationary vortex-shedding effects in non-synoptic wind response modeling of tall buildings [20] demonstrated the relative contribution of vortex-shedding loads to cross-wind response under downburst conditions. Uncertainty propagation from uncertain models of thunderstorm wind speed [26] used interval analysis to quantify the response upper and lower bounds arising from epistemic uncertainty. Dynamic response of a flat-roof building under a nonstationary moving microburst [89] used time-domain integration to compute roof nodal displacements.

Hybrid simulation of thunderstorm outflows and structural wind-excited response [111] spliced numerical simulation with field measurements to generate a hybrid wind field. Research on longitudinal forces in transmission towers caused by asymmetric downburst conductor loads [30] identified the conditions of critical angle of attack at 30 degrees and ratio of radial distance to jet diameter of 1.6. Research on dynamic effects of downburst wind on longitudinal forces in transmission towers [7] found that longitudinal reactions can reach 60 percent of transverse-wind reactions.

7 Structural-Archetype Evidence Comparison

7.1 Tall buildings, low-rise buildings, roofs, and cladding

The classification of structural archetypes serves an organizational function in transient wind-effect research. Differences in dynamic characteristics, load sensitivity, and response metrics across archetypes mean that direct cross-archetype quantitative comparison lacks a physical basis. The fundamental period of tall buildings is typically on the order of several seconds, with their response controlled by low-frequency load components; the fundamental period of low-rise buildings is far shorter than the time-varying characteristic time of the load, making their response approximately quasi-static; tower-conductor systems exhibit unique system-coupling features due to the nonlinear geometric stiffness of conductors and the non-uniformity of loads over long spans; long-span structures and bridges have closely spaced modes, and the simultaneous participation of multiple modes introduces additional complexity for modal truncation and equivalent-static-force construction. These differences require that response evidence be organized by archetype.

Response observables for building-type structures can be divided into four levels by physical standing: global displacement and acceleration, base internal forces, member demands, and cladding-surface load demands. Research on transient loads on buildings in microbursts and tornadoes [5] recorded the ratio of force coefficients to quasi-static references on cube and tall-building models. Large-eddy simulation of microburst wind flow around buildings [15] resolved the downburst flow field down to building scale through nested grids. The four levels of observables are not interchangeable.

Along-wind response analysis of tall buildings under transient nonstationary wind [59] treated nonstationary equivalent static demands from framework and case perspectives. An experimental study of wind-induced loads on tall buildings under downburst outflow conditions [29] measured the base-shear dynamic amplification factor for the CAARC model under three outflow environments in a multi-fan wind tunnel. Nonstationary vortex-shedding effects in non-synoptic wind response modeling of tall buildings [20] demonstrated the relative contribution of vortex-shedding loads. Parametric study of the along-wind response of the CAARC building in downbursts [115] demonstrated, by varying wind-field parameters, the sensitivity of peak response to downburst size and translation velocity.

Research on roof aerodynamic forces on buildings of different heights in transient wind [145] recorded the interaction effect between roof height and transient wind profile. Inelastic response of base-isolated tall buildings under nonstationary wind [78] provided probabilistic descriptions of response quantities on a nonlinear path.

7.2 Transmission towers, conductors, guyed towers, and line systems

The modeling level of the tower-conductor coupled system determines the physical meaning of the response. Research on the hazard of downburst formation and dissipation to transmission towers [1] provided a comprehensive organization of wind-field characteristics at each life-cycle stage of the downburst. Research on dynamic effects of downburst wind on longitudinal forces in transmission towers [7] found in a six-span continuous model that longitudinal reactions can reach 60 percent of transverse-wind reactions. Research on longitudinal forces in transmission towers caused by asymmetric downburst conductor loads [30] provided simplified estimation charts for longitudinal forces based on dimensionless parameters.

Dynamic response analysis of transmission towers under downburst wind-wave loads [66] considered the coupled input of wind and waves in an offshore environment. Full-scale downburst wind loads measured on a transmission line in northern Germany [146] provided one of the few full-scale records available. Critical load cases for lattice transmission-line structures under downburst action [49] discussed the controlling relationship of different downburst positions on structural demands. Probabilistic study of nonstationary extreme-value response of transmission towers under moving downburst impact [47] discussed, within a numerical simulation framework, the constraint that limited event samples impose on extreme-value estimation.

7.3 Cooling towers, shells, long-span structures, bridges, and wind turbines

The response of cooling towers and thin-shell structures is primarily controlled by load circumferential distribution, shell curvature, and local stability. Research on collapse mechanisms and failure criteria of super-large cooling towers subjected to downbursts [73] demonstrated that different radial positions can produce different failure mechanisms. Research on pressures on a large-span hyperbolic spherical-shell roof [77] showed that terrain and radial position alter pressure zoning.

An analytical load model for bridge decks under downbursts [50] mapped radial-velocity profiles to span-wise distributions. Comparison of transient response of bridges under downbursts and typhoons [95] showed that the two input types lead to different response envelopes. Research on quasi-static parameters of wind turbines under downburst conditions [38] obtained the maximum tower-base bending moment at a source-specific radial position. A fluid-structure-interaction scaled model [52] recorded the bidirectional interaction between aerodynamics and structural motion. Dynamic analysis of a horizontal-axis wind turbine [126] reported time-history response under parked conditions.

Research on multi-hazard vulnerability surfaces of suspension bridges [10] incorporated pre-existing damage states and microburst input together into the suspension bridge model. The sparse identification method [92] identified time-varying aerodynamic forces from bridge response data. A time-frequency simulation study [40] compared displacement and acceleration from measured and parameterized inputs.

7.4 Cross-archetype comparable and non-transferable quantities

Table 4 . Structural archetypes, input objects, response metrics, and limitations

Structural archetype	Typical input objects	Primary response metrics	Primary limitations	Representative sources
Tall buildings	Wind-speed profile, pressure field	Roof-top displacement, acceleration, base forces	Nose-shaped profile affects modal-load allocation	[59] [29] [20]
Low-rise buildings	Pressure field, force coefficients	Roof suction, net forces	Separated-flow reattachment, area averaging	[5] [36] [145]
Transmission tower-conductor system	Wind-speed field, force coefficients	Support reactions, conductor tension, longitudinal force	Multi-span coupling, geometric nonlinearity	[1] [7] [30] [66]
Cooling towers and shells	Pressure field	Shell internal forces, buckling	Load circumferential distribution, local stability	[73] [77]
Bridges	Wind-speed field, aerodynamic force coefficients	Displacement, internal forces	Closely spaced modes, span-wise correlation	[50] [95] [84]
Wind turbines	Wind-speed field, blade aerodynamic forces	Tower-base bending moment, blade-root bending moment	Operating state, control strategy	[38] [52] [126]
Lighting poles and slender structures	Wind-speed field	Tip displacement, DAF	Low damping, single-mode dominance	[86] [54] [49]
CAARC benchmark building	Wind speed or pressure field	Base forces, roof-top displacement	Specific geometry and damping	[115] [33] [91]
Long-span roofs	Pressure field	Nodal displacement, internal forces	Terrain, sloped-ground effects	[14] [119]
Method-verification SDOF/MDOF	Evolutionary spectrum or time history	Response statistics	Idealized structures	[8] [128] [13]

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Structural archetype	Typical input objects	Primary response metrics	Primary limitations	Representative sources
Suspension bridges	Seismic and microburst	Multi-hazard vulnerability	Limit-state definition	[10] [92]

Note: Cells are qualitative synthesis of source-local or method attributes and do not support cross-source quantitative derivation.

The cross-archetype transferable components are primarily mathematical components such as the equation of motion, numerical integration, reduced-order modeling, and uncertainty representation; load mapping, response-metric thresholds, and numerical magnitudes must be re-established on the target archetype. Parameterization of downburst loads in industry [98] can be used to identify input-variable categories, but its applicable object does not constitute cross-industry response criteria.

8 Response Metrics and Evaluation Quantities

8.1 Displacement, acceleration, internal force, tension, and local demands

The definitions of peak and root-mean-square values depend on the choice of time window; under transient-input conditions, the choice of window length itself changes the physical meaning of the statistics. A directionality response study of thunderstorm outflows [22] compared peak-value differences between directional-tracking and fixed-direction methods on two structures with different cross-sections. Wavelet and transform singular-value-decomposition system-identification methods for nonstationary response [37] provided a framework for extracting time-varying modal parameters from transient response.

In a nonstationary process, the meaning of the root-mean-square changes with the statistical window length: when the window length is shorter than the characteristic time scale of the modulation function, the RMS value tracks the envelope of transient peaks; when the window length spans the entire event duration, transient peaks are smoothed by the low-amplitude phases of the entire process. This window-length dependence means that RMS values reported by different sources cannot be directly compared without first verifying the window-length definition. Peak extraction is also affected by the window-function shape and overlap ratio: a rectangular window introduces spectral leakage at window boundaries, while a cosine-tapered window reduces leakage but degrades peak-amplitude fidelity for transient events in the time domain. For thunderstorm outflow events with durations of tens of seconds to several minutes, window-length selection requires a trade-off between time resolution and statistical stability: windows that are too short lead to excessively wide confidence intervals for variance estimates, while windows that are too long average out nonstationary features. When the analysis object extends from a single degree of freedom to a multi-degree-of-freedom system, different modal components have different sensitivities to window length: higher-mode response components, having higher frequencies, are more sensitive to short windows, while the fundamental mode is more influenced by long-period modulation.

8.2 Peak, RMS, peak factor, DAF, MDMF, and response-spectrum quantities

The definitions of the dynamic amplification factor and the maximum dynamic magnification factor involve ambiguity in the reference static state. Research on practical methods for digital simulation of non-synoptic wind speeds and their implications for single-pole tower response [60] provided a simplified path for estimating single-pole tower response under nonstationary conditions. A field study of wind and structural response of lighting poles in downbursts [86] extracted the dynamic amplification factor for lighting poles from full-scale records. Hybrid simulation of thunderstorm outflows and structural wind-excited response [111] applied dynamic amplification to the response

from a hybrid wind-field basis. Data-driven and model-based perspectives on gust-front factors [93] discussed the difference between the gust-factor definition under nonstationary conditions and the classical gust-load factor. The denominator definitions of the dynamic amplification factor differ among the above sources; therefore, their numerical values cannot be directly juxtaposed.

8.3 Vulnerability, first-passage probability, residual response, and damage indices

The applicability of the peak factor and first-passage probability degrades under nonstationary input. Research on wind-induced vibration and control of string structures under thunderstorm downbursts [135] recorded the variation of nodal displacement peak factors for a string dome under downburst time-history input. Sensitivity analysis of structural response to thunderstorm downburst models [46] demonstrated, on two slender structures, the sensitivity of the peak factor to wind-field model parameters. Research on response and first-passage probability of a linear-elastic SDOF system under nonstationary random excitation based on the S-transform [144] provided a computational path for first-passage probability in the time-frequency domain. Peak-factor and gust-factor estimation for nonstationary wind speeds [132] proposed a corrected peak-factor definition applicable to nonstationary processes.

Table 5 . Differences in response-metric definitions and comparability obstacles

Metric category	Definition dependencies	Comparability obstacles	Representative sources
Peak displacement	Time window, reference displacement	Different window lengths produce different peak meanings	[22] [115]
Root-mean-square	Statistical window length	Short windows track the envelope; long windows smooth transients	[37] [41]
Dynamic amplification factor	Reference static state	Denominator definition not unified	[60] [86] [111]
Peak factor	Time window, mean definition	Systematic bias in nonstationary processes	[132] [46] [135]
First-passage probability	Crossing-rate computation method	Poisson assumption degrades under nonstationarity	[144] [9]
Equivalent static demand	Target response quantity	Target change causes equivalent distribution change	[58] [59] [131]
Vulnerability	Conditioning variable	Curves from different conditioning are not interchangeable	[79] [10]
Residual response	Damage state	Not interchangeable with elastic response	[142] [113]

Note: Cells are qualitative synthesis of source-local or method attributes and do not support cross-source quantitative derivation.

8.4 Literature-comparison limitations arising from metric-definition differences

The construction of equivalent static demands as a metric is subject to dual limitations from the target response quantity and the reference state. Hierarchical evaluation of LES for tall-building loads and response [23] treated mean, background, and resonance errors separately. The generalized gust-front factor and along-wind analysis of tall buildings [58] [59] constructed nonstationary equivalent load distributions based on target response. Inelastic response of base-isolated buildings [78] showed that when stiffness and energy dissipation vary with state, the same static distribution cannot remain equivalent throughout the time history.

The meaning of vulnerability and exceedance probability is determined by the choice of conditioning variable. Research on dynamic response of slender vertical structures under thunderstorm outflows [114] used a frequency-domain method to estimate peak displacement. Research on dynamic response of slender structures under thunderstorm outflow excitation [125] provided measured-versus-numerical comparisons. Research on maximum dynamic response based on evolutionary-spectral models [128] provided analytical estimates of maximum response from evolutionary-spectral input.

Residual response and member internal forces are damage-state quantities whose physical meaning is not interchangeable with elastic response quantities. Coupled dynamic response of a tall building in thunderstorm downburst simulations [33] recorded the bimodal time-history characteristics of non-turbulent downburst velocity on the CAARC model. Dynamic response of a flat-roof building under a nonstationary moving microburst [89] used time-domain integration to compute roof nodal displacements.

Research on the response of linear systems under nonstationary random filtered input [8] provided analytical second-order-moment solutions within an elastic framework. Peak-factor and gust-factor estimation for nonstationary wind speeds [132] noted that the peak factor systematically biases peak estimates in nonstationary processes. Research on the influence of nonstationary wind-speed models on tall-building response under thunderstorm outflows [41] and representation of nonstationary random processes under nonstationary wind [111] respectively treated response-metric uncertainty from different perspectives.

The influence of nonstationary wind-speed models on response metrics manifests differently on different structures. Sensitivity analysis of slender structures [46] confirmed that the magnitude of the effect of damping-ratio variation on peak response is of the same order as the uncertainty in wind-field parameters. Peak-factor and gust-factor estimation for nonstationary wind speeds [132], roof aerodynamic forces on buildings of different heights [145], gust-front factor framework [131], CAARC response parametric study [115], and nonstationary-random-process characterization [114] together show that identical metric names are not sufficient to support numerical merging.

Input states in bridges and wind turbines likewise cannot be compressed into a unified multiplier. Long-span bridge surrogate model [68], the fluid-structure coupling path [90], and comparison of downbursts versus typhoons [95] respectively retain varying degrees of span-wise distribution and time-frequency information. Bridge-deck nonstationary wind-field decomposition [141] was used to extract spatial-correlation structure. Wind-turbine parametric study [38] and time-frequency model [40] distinguished among radial position, operating state, and event input. Bridge-deck transient aerodynamic forces [84] additionally include flow-history effects.

Equivalent representations in tower-conductor systems must preserve the force transfer between conductors and tower body. Frequency-domain analysis of transmission towers [35] constructed equivalent demands from background and resonant responses. Tower-body background-resonance separation [34] and conductor-response research [61] respectively adopted different state quantities. Quasi-static buffeting analysis of transmission lines [85] and nonlinear analytical prediction [27] further showed that moving load position and the proportion of the mean component alter the spatial distribution of equivalent forces. When the structure already has local damage or the loads involve extreme-value probability, the aeroelastic response of partially damaged towers [25] and moving-downburst extreme-value studies [47] provide conditional response evidence.

Nonlinear buildings and spatial structures require understanding equivalent demands in a state-dependent manner. Cross-wind aeroelastic response of tall buildings [87] and reduced-order stochastic analysis of slender buildings [107] respectively treated motion-dependent aerodynamic forces and nonlinear state evolution. Collapse analysis of string structures [120] incorporated cable-force loss and failure paths into the response computation. Nonstationary response of tall buildings [41] and vortex-shedding effect studies [20] showed that, even when the structure remains elastic, variations in the input frequency band and aerodynamic-force composition cause different time segments to correspond to different controlling responses.

Dynamic response analysis of transmission towers under downburst wind-wave loads [66] included an additional wave-contribution component in offshore-condition response quantities. Dynamic response of conductors during transmission-tower collapse [142] showed that after the tower body enters the collapse stage, the time-history characteristics of conductor tension undergo a fundamental change.

9 Uncertainty Propagation and Evidence Strength

9.1 Wind-field model uncertainty

Uncertainty in inflow characterization is the starting link of the response-uncertainty chain. Research on simulating fully nonstationary random processes using generalized harmonic wavelets [31] demonstrated the evolutionary-spectral convergence accuracy at 1,000 samples for nonstationary-process sample generation. A stochastic harmonic-function representation of nonstationary random processes [51] provided an alternative path for generating nonstationary realizations from the frequency-domain perspective. The evolutionary-spectral models on which both methods depend originate from limited field records or parametric assumptions; therefore, model selection in inflow characterization is the first source of uncertainty.

9.2 Pressure/force conversion and spatial-mapping uncertainty

Uncertainty in pressure normalization propagates to the response through the choice of reference wind speed and reference height. A deterministic-stochastic hybrid model for downbursts [99] partitioned wind-field parameters into a deterministic envelope and stochastic fluctuations; the choice of reference wind speed directly affects the relative weights of the two parts. Wind-tunnel measurement of non-Gaussian and nonstationary aerodynamic features of a rectangular cross-section [69] recorded the influence of reference-dynamic-pressure selection on the meaning of the pressure coefficient.

Integration schemes and correlation assumptions in load mapping introduce systematic bias. Research on uncertainty propagation from uncertain models of thunderstorm wind speed [26] used interval analysis to quantify the response upper and lower bounds arising from epistemic uncertainty, with turbulence intensity being the parameter with the largest influence on response spread. Research on wind pressure and wind-field response of a large-span hyperbolic-paraboloid shell roof under downbursts [77] demonstrated the systematic influence of grid resolution and terrain-model selection on pressure distributions in numerical load mapping.

9.3 Structural parameters, damping, frequency, and nonlinear model uncertainty

The uncertainty of damping and frequency has a prominent effect on the amplification of resonant components under transient input. Research on time-varying structural reliability under nonstationary non-Gaussian processes [9] calibrated the influence of damping-parameter uncertainty on exceedance probability in crossing-rate calculations. Stochastic analysis of vertical-structure vulnerability under downburst wind [79] treated structural frequency and damping ratio as random variables; the variability of these two parameters has a significant broadening effect on the vulnerability curve. Sensitivity analysis of structural response to thunderstorm downburst models [46] confirmed, on two slender structures, that the magnitude of the change in peak response when the damping ratio varies from 0.2 percent to 0.5 percent is of the same order as the uncertainty in wind-field parameters.

The uncertainty of nonlinear models exhibits path-dependent characteristics. Research on multi-hazard vulnerability surfaces of suspension bridges [10] showed that the choice of constitutive model and failure criterion alters the shape of the vulnerability surface. The reduced-order method for coupled nonlinear stochastic response of slender buildings [107] obtained different response statistics under different damping models.

9.4 Sample representativeness, validation data, and statistical extrapolation limitations

Filtering, sampling, and peak-identification methods in response extraction constitute technical-level uncertainty sources. Research on first-passage probability under nonstationary excitation based on the S-transform [144] discussed the influence of time-frequency resolution on crossing-rate estimation. Modeling of vortex-shedding effects in nonstationary wind on tall buildings [20] used specific filtering parameters during peak identification; different filter settings produce different peak sequences.

Sample representativeness is the terminal constraint of the uncertainty chain. Single events, single configurations, and limited repetition counts together constrain the reliability of statistical inference. A probabilistic perspective on numerical simulation of wind effects [65] discussed, at the methodological level, the influence of limited samples on extreme-value estimation. Research on three-dimensional nonstationary non-Gaussian wind fields [109] emphasized the scarcity of measured records simultaneously satisfying three-dimensionality, nonstationarity, and non-Gaussianity. Probabilistic study of nonstationary extreme-value response of transmission towers under moving downburst impact [47] assessed the uncertainty in extreme-value distribution parameters under limited event numbers. Insufficient sample representativeness is especially acute in extreme-value estimation: the shape parameter of generalized extreme-value or generalized Pareto distributions is highly sensitive to sample size, and when the number of available events is fewer than several tens, the confidence interval for the shape parameter spans both positive and negative tail behaviors, at which point the choice of distribution type may significantly alter extrapolation results. Monte Carlo methods compensate for limited observational samples through large numbers of random realizations, but the premise is that the random input model itself captures the key physical features, including nonstationary modulation, spatial coherence, and non-Gaussianity. When the parameters of the random model are calibrated from limited field records, parameter uncertainty propagates to the generated samples, widening the scatter range of response statistics. Although Latin hypercube and importance sampling methods can reduce the sample count needed to achieve a given confidence level, the efficiency gains of these acceleration methods under high-dimensional and strongly nonlinear conditions depend on the accuracy of the prior knowledge.

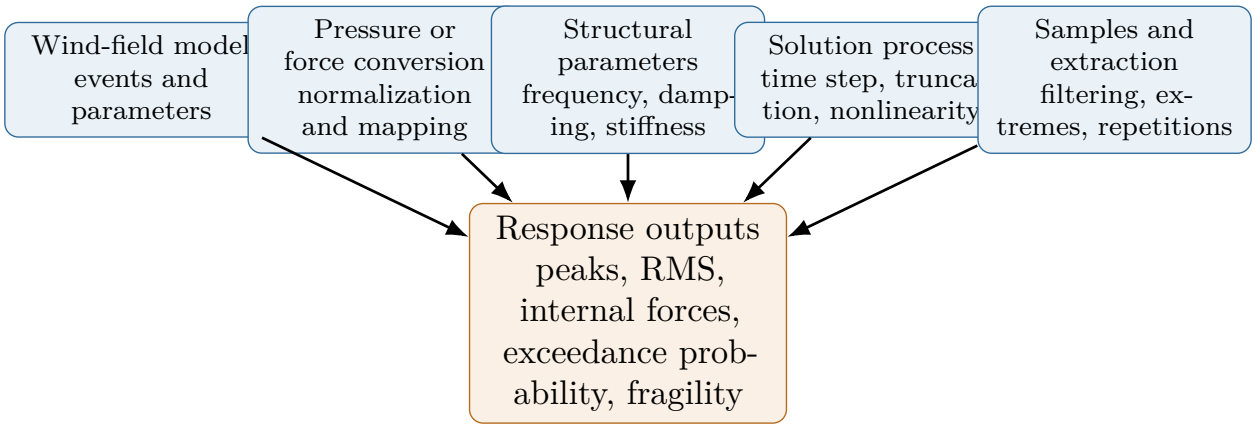


Figure 5 Uncertainty propagation path diagram. Original synthesis based on exact146 accepted-only evidence; does not redraw source numerical curves and does not represent a PETW computation result. From left to right, five uncertainty source layers are shown: wind-field characterization (event samples, model selection, parametric assumptions), conversion (pressure normalization, integration schemes, correlation assumptions), structural parameters (frequency, damping, mass, stiffness), solution (time step, modal truncation, nonlinear path), and sampling (limited repetitions, extreme-value extraction, filter settings). Uncertainty from each layer propagates to the response output (peak, RMS, exceedance probability, vulnerability). Interaction-term markers indicate that coupling effects may exist between layers.

Table 6 . Uncertainty sources and affected outputs

Uncertainty layer	Typical sources	Affected outputs	Source keys
Wind-field characterization	Limited event samples, model parameterization	Load envelope, frequency content	[31] [51] [99]
Pressure normalization	Reference wind speed, reference height	Force-coefficient meaning	[69] [99]
Load mapping	Integration scheme, correlation model	Nodal forces, generalized forces	[26] [77]
Damping and frequency	Measurement uncertainty, nonlinear degradation	Resonant-component amplification	[9] [79] [46]
Nonlinear path	Constitutive model, failure criterion	Vulnerability surface shape	[10] [107]
Response extraction	Filtering, sampling, peak identification	Peak sequence	[144] [20]
Sample representativeness	Event count, configuration count	Reliability of statistical inference	[65] [109] [47]

Note: Cells are qualitative synthesis of source-local or method attributes and do not support cross-source quantitative derivation.

Consequently, uncertainty propagation should not be compressed into a single safety factor. For the same structure, the scale and moving path of the input wind field alter the low-frequency load envelope; the spatial coherence model alters the phase combination of generalized forces; uncertainty in damping and frequency alters the resonant component; and the metric-extraction rules further affect the numerical values of peaks and exceedance probability. Only when load normalization, spatial mapping, structural dynamic parameters, and response-extraction criteria are simultaneously aligned does cross-method uncertainty comparison have a definite meaning [2] [26] [9].

10 Interface Handoffs with WindField, WindTunnel/CFD, and DesignLoad

10.1 Input-requirement write-back to WindField

The handoff object of Step 4c is composed jointly of response evidence, metric definitions, sensitivity information, and use limitations. Research on thunderstorm-wind design parameters [96] distinguished, from the downstream-demand end, the categories of information that response analysis must provide; the performance-based path relying on historical records and Monte Carlo simulation [105] showed that subsequent parameter generation depends on whether evidence conditions and uncertainty descriptions are complete. Accordingly, each piece of response evidence must simultaneously carry its structural configuration, load representation, response-quantity type, normalization baseline, validation object, and transfer limitation.

10.2 Pressure/force output specification requirements for WindTunnel/CFD

This handoff does not include design-use coefficients, dimensionless design multipliers, code-applicability judgments, or engineering adoption conclusions. Evidence from a study of critical load cases for transmission-line structures [49] serves to distinguish the source-local relationship between different downburst positions and structural demands. The applicability scope of a study on parameterization of downburst loads for wind turbines [98] is tied to a specific turbine type and parked state. Both types of sources can indicate which variables and response quantities need to be preserved, but a literature review alone cannot complete the conversion from analytical evidence to design values.

A verification study of standard CFD setups for wind-load assessment of tall buildings [91] provided an exemplar of a four-stage verification chain for the LES workflow on the CAARC benchmark, but its verification scope is confined to specific grids, subgrid models, and reference wind-tunnel data. A downburst loading assessment [16] organized the hierarchical relationships among wind-field generation, load application, and structural response from the transmission-line perspective. A review of transmission-line systems under downburst loads [56] organized the coverage of existing research in load generation, structural modeling, and response analysis.

10.3 Response evidence, sensitivity, and use limitations for DesignLoad

The handoff preconditions include three items of information that must be transferred together with the evidence. First, comparability conditions: response quantities reported by different sources can be placed in the same comparison framework only when the input definition, structural model, normalization baseline, and validation method are consistent across all four dimensions. Second, archetype consistency: quantitative transfer across archetypes has not been systematically validated in the existing evidence. Third, uncertainty declaration: the variability of the inflow model, structural parameters, and solution method on which each piece of response evidence relies must be explicitly tagged at the time of handoff.

Research on thunderstorm-wind design parameters [96] and the performance-based design method [105] together showed that the conversion from response evidence to design parameters requires independent reliability targets and human review. The CAARC verification study [91] demonstrated the organization of a verification chain, but its scope is confined to a specific configuration. The transmission-line review [56] and the transmission-technology assessment [16] organized the coverage in load generation and structural modeling for tower-conductor systems. Wind-turbine load parameterization [98] and transmission-tower critical cases [49] respectively tagged, from the wind-turbine and tower-conductor system perspectives, the variable categories to be transferred.

10.4 Reasons the evidence cannot yet be upgraded to design conclusions

The existing accepted-only evidence can delineate method categories, input objects, response metrics, and source-local limitations, but has not yet produced a PETW-specific structural response computation, cross-archetype validation, unified normalization baseline, or approved uncertainty-propagation result. The structural geometries, event definitions, pressure baselines, damping models, and response metrics from different sources have not been aligned; therefore, aggregating these results into unified coefficients would exceed comparability boundaries.

From response evidence to design conclusions, independent hazard definitions, target reliability, load combinations, archetype applicability, and human scientific review are still required. This review does not perform these steps, does not enter design-value derivation, does not derive design-use coefficients from local evidence, and does not convert frameworks or parameters from the sources into PETW adoption values. Scientific status remains `not_approved`; formal manuscript status remains `not_applied`.

Table 7 . Interface requirements and limitations across WindField, WindTunnel/CFD, Structural, and DesignLoad

Interface	Information transferred	Required accompanying conditions	Items not included	Representative sources
WindField to Structural	Wind-speed field, evolutionary spectra, time-varying coherence	Event type, station, profile model	Climate statistics or return periods	[2] [4] [109]

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Interface	Information transferred	Required accompanying conditions	Items not included	Representative sources
WindTunnel/CFD to Structural	Pressure field, force coefficients, base forces	Scale ratio, grid, turbulence model, normalization baseline	Cross-apparatus extrapolation	[3] [23] [91]
Structural to DesignLoad	Response evidence, metric definitions, sensitivity	Structural configuration, load type, validation scope, uncertainty tags	Design coefficients, code-adoption conclusions	[96] [105] [98]

Note: Cells are qualitative synthesis of source-local or method attributes and do not support cross-source quantitative derivation. This table is an evidence-field registry, not a design or code-adoption conclusion.

Several unresolved questions remain in the existing evidence. Whether the interface definition between background and resonant components under nonstationary conditions changes with modulation rate, and at what modulation rate the classical decomposition completely fails, have been explored locally in the existing literature but no consistent criterion has been established. Most archetype-level evidence comes from single or few event configurations, and the reliability of statistical extrapolation has not been systematically assessed. The quantitative difference between path-dependent responses of nonlinear structures under transient input and linear analysis results remains sparsely covered on building and bridge archetypes. Resolution of these questions requires new field data and systematic cross-research-group comparisons.

Literature sources: The reference list is generated by post-processing from 146 frozen BibTeX entries and is not repeated here.

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